

# TEST CASE PLAN

## AlgoNova – AI-Powered Intelligent Learning Platform

|                    |                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Project            | AlgoNova                                                                                                                                |
| Document           | Software Testing – Test Case Plan                                                                                                       |
| Version            | 1.0                                                                                                                                     |
| Application Type   | Web-based AI-powered educational platform                                                                                               |
| Technology Context | React, Node.js, Express.js, MongoDB Atlas, Google Gemini AI                                                                             |
| Primary Modules    | Authentication, Dashboard, Learning, DSA Visualizer, Coding Practice, AI Assistant, Mock Interview, System Design Sandbox, Gamification |

### 1. Purpose

This test case plan defines the testing approach for AlgoNova and identifies functional, integration, usability, security, performance, and AI-related scenarios to be validated before release. The objective is to ensure that the platform works correctly, protects user data, and provides reliable learning functionality and feedback across supported modules.

### 2. Testing Objectives

- Verify that all major features meet their specified requirements.
- Validate communication between the React frontend, Node.js/Express backend, MongoDB Atlas, and Gemini AI services.
- Identify functional defects, integration issues, validation errors, and usability problems.
- Verify authentication, authorization, session handling, and protection of user data.
- Validate responsive behavior across supported screen sizes.
- Check that AI-generated responses are relevant, consistent, and appropriately constrained.
- Ensure user progress, XP, badges, streaks, and assessment results are stored and displayed correctly.

### 3. Testing Levels and Types

| Testing Type            | Purpose                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| Unit Testing            | Test individual components, functions, API handlers, and validation logic.                               |
| Integration Testing     | Verify communication among frontend, backend, database, authentication, and AI services.                 |
| System Testing          | Validate complete end-to-end user workflows.                                                             |
| UI / Usability Testing  | Check navigation, layout, readability, responsiveness, and interaction.                                  |
| Security Testing        | Validate authentication, authorization, input validation, session security, and API protection.          |
| Performance Testing     | Check response time, concurrent requests, API behavior, and page performance.                            |
| AI Response Testing     | Evaluate relevance, correctness, consistency, hallucination risks, and handling of inappropriate inputs. |
| User Acceptance Testing | Confirm that the platform satisfies intended student-learning requirements.                              |

## 4. Detailed Test Cases

All test cases below are marked Passed as the current documented test outcome.

| TC ID  | Module         | Test Scenario                   | Test Steps / Input                                   | Expected Result                                                                                | Priority | Status |
|--------|----------------|---------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------|----------|--------|
| TC-001 | Registration   | Register with valid details     | Enter valid name, email and password; submit.        | Account is created and user is redirected appropriately.                                       | High     | Passed |
| TC-002 | Registration   | Reject invalid email            | Enter invalid email format and submit.               | Validation message is displayed; account is not created.                                       | High     | Passed |
| TC-003 | Login          | Login with valid credentials    | Enter registered credentials.                        | User is authenticated and dashboard opens.                                                     | High     | Passed |
| TC-004 | Login          | Reject incorrect password       | Enter valid email with incorrect password.           | Login fails with an appropriate error message.                                                 | High     | Passed |
| TC-005 | Authentication | Unauthorized protected route    | Open protected page without authentication.          | User is redirected to login or access is denied.                                               | High     | Passed |
| TC-006 | Dashboard      | Load user dashboard             | Login and open dashboard.                            | Progress, XP, streak, recommendations and statistics load correctly.                           | High     | Passed |
| TC-007 | Learning       | Open a learning topic           | Select a DSA topic from the learning module.         | Correct topic content loads without broken UI.                                                 | Medium   | Passed |
| TC-008 | Visualizer     | Run algorithm visualization     | Select an algorithm and start visualization.         | Algorithm executes visually step-by-step with correct state transitions.                       | High     | Passed |
| TC-009 | Visualizer     | Pause and resume visualization  | Start visualization; pause; resume.                  | Animation pauses and continues from the correct state.                                         | Medium   | Passed |
| TC-010 | Coding         | Submit valid solution           | Submit code for a supported coding problem.          | Code is processed and result/status is displayed correctly.                                    | High     | Passed |
| TC-011 | Coding         | Handle invalid code             | Submit syntactically incorrect code.                 | System reports an appropriate error without crashing.                                          | High     | Passed |
| TC-012 | AI Assistant   | Ask technical question          | Enter a DSA-related question.                        | AI returns a relevant educational explanation.                                                 | High     | Passed |
| TC-013 | AI Assistant   | Request code explanation        | Provide code and request an explanation.             | AI explains the logic and relevant complexity appropriately.                                   | High     | Passed |
| TC-014 | AI Review      | Review submitted solution       | Submit a coding solution for review.                 | Feedback identifies correctness, complexity, and possible improvements where applicable.       | High     | Passed |
| TC-015 | Mock Interview | Start interview session         | Select interview type and start.                     | Interview session begins and questions are presented.                                          | High     | Passed |
| TC-016 | Mock Interview | Submit interview response       | Answer an interview question and continue.           | Response is recorded and the next step proceeds correctly.                                     | High     | Passed |
| TC-017 | Mock Interview | Generate interview feedback     | Complete an interview session.                       | Score/feedback and improvement areas are displayed.                                            | High     | Passed |
| TC-018 | System Design  | Create architecture             | Add components and connect them in the sandbox.      | Components and relationships are rendered correctly.                                           | High     | Passed |
| TC-019 | System Design  | Save design                     | Create a design and select save.                     | Design is persisted and can be retrieved by the user.                                          | Medium   | Passed |
| TC-020 | Gamification   | Award XP after activity         | Complete an eligible learning activity.              | XP is updated according to configured rules.                                                   | Medium   | Passed |
| TC-021 | Gamification   | Update streak                   | Complete activity on an eligible consecutive day.    | Learning streak is updated correctly.                                                          | Medium   | Passed |
| TC-022 | Progress       | Record assessment result        | Complete a quiz or assessment.                       | Score and completion status are stored and shown correctly.                                    | High     | Passed |
| TC-023 | Database       | Persist user progress           | Complete activity, then refresh or re-login.         | Previously stored progress remains available.                                                  | High     | Passed |
| TC-024 | Security       | Reject unauthorized API request | Call protected API without valid authentication.     | API denies access and does not expose protected data.                                          | High     | Passed |
| TC-025 | Responsive UI  | Test mobile layout              | Open key pages at a mobile viewport.                 | Content remains usable without major overlap or horizontal overflow.                           | Medium   | Passed |
| TC-026 | Error Handling | Backend/API unavailable         | Simulate API failure while loading a feature.        | User receives a meaningful error state and UI does not crash.                                  | High     | Passed |
| TC-027 | AI Safety      | Unsupported or malicious prompt | Send an unsafe or off-topic prompt to the AI module. | System handles the prompt according to configured safety behavior and does not expose secrets. | High     | Passed |
| TC-028 | Logout         | Logout from active session      | Click logout and revisit a protected page.           | Session is terminated and protected content is inaccessible.                                   | High     | Passed |

## 5. Test Environment

| Component  | Test Environment                                                                                   |
|------------|----------------------------------------------------------------------------------------------------|
| Client     | Modern Chromium/Firefox/Safari browser; desktop and mobile viewport testing.                       |
| Frontend   | React-based web application.                                                                       |
| Backend    | Node.js + Express.js API.                                                                          |
| Database   | MongoDB Atlas test database.                                                                       |
| AI Service | Google Gemini API test configuration.                                                              |
| Network    | Normal broadband/mobile network plus simulated slow or unavailable API conditions.                 |
| Test Data  | Dummy student accounts, sample DSA problems, sample code, interview responses, and system designs. |

## 6. Entry and Exit Criteria

| Entry Criteria                                             | Exit Criteria                                                 |
|------------------------------------------------------------|---------------------------------------------------------------|
| Requirements and major workflows are defined.              | All critical and high-priority test cases have been executed. |
| Test environment and test data are available.              | No open Critical defects remain.                              |
| Build is deployable and major dependencies are configured. | High-severity defects are fixed or formally accepted.         |
| Required APIs and database connections are available.      | Regression testing is completed for affected modules.         |
| Test cases are reviewed before execution.                  | Test results and unresolved issues are documented.            |

## 7. Defect Classification

| Severity | Meaning                                                                        |
|----------|--------------------------------------------------------------------------------|
| Critical | System crash, major security failure, data loss, or core application unusable. |
| High     | Major feature does not work or an important workflow is blocked.               |
| Medium   | Feature works with a significant issue or a workaround is available.           |
| Low      | Minor UI, wording, alignment, or non-critical usability issue.                 |

## 8. Test Summary

All 28 documented test cases in this test plan are marked as Passed. The test cases cover authentication, learning, DSA visualization, coding practice, AI assistance, mock interviews, System Design, gamification, progress persistence, security, responsive UI, error handling, and logout.

## 9. Conclusion

The test case plan provides structured coverage for AlgoNova's major functional and non-functional requirements. Testing should be performed incrementally throughout development, with particular attention to authentication, learner-progress persistence, AI integrations, coding workflows, System Design interactions, and error handling.